

Northeastern University

Report for Experiment #1

Measurement

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Experiment Date: Jan 14, 2025

Abstract

This experiment focused on learning measurement techniques and handling errors through two investigations: figuring out the density of metal cylinders (Investigation 1) and studying the statistical variation in background radiation (Investigation 2). We analyzed uncertainties and variations, which gave us consistent density results and a bell-shaped histogram for the radiation counts.

Introduction

The precision of measurements and understanding of uncertainties are foundational to experimental physics. This experiment involved:

Investigation 1: Measurement of mass, length, and diameter of metal cylinders to calculate density and uncertainties.

Investigation 2: Statistical variation in background radiation counts using a Geiger counter.

The experiment highlighted the importance of error propagation, statistical distributions, and data visualization.

Investigation 1

Objective: To calculate the density of metal cylinders and analyze the associated uncertainties.

Method: Mass, length, and diameter of four cylinders were measured.

Volume (V) was calculated as: $V = \pi LD\text{^}2/4$

Density (ρ) was determined using: $\rho = /V$

Uncertainties in V and ρ were propagated.

Results:

Average Density: 8.48 g/cm^3

Standard Error: 0.09 g/cm^3

Graph: A linear Mass vs. Volume graph was plotted: y = 0.1192x−0.041

Investigation 2

Objective: To analyze statistical variations in background radiation counts.

Method:

Radiation counts per minute were recorded for 60 trials using a Geiger counter.

A histogram was plotted, and key statistical measures were computed:

Mean: 18.17 counts/60s.

Standard Deviation: 4.37.

Uncertainty in Mean: 0.56.

Histogram-Based Uncertainty: 4.25.

Full Width at Half Maximum (FWHM) was used to estimate histogram-based uncertainty.

Graph:

The histogram showed a normal distribution with a mean of 18.17 counts/60s.

Conclusion

This experiment focused on the importance of precise measurements and understanding errors. In Investigation 1, we got consistent density values for the cylinders, showing how reliable our measurement methods were. Investigation 2 showed how random radioactive decay is, with a bell-shaped histogram that made the statistical variation clear.

Questions

1. Effect: Mass measurements would include a constant offset.
   Error Type: Systematic error.

2. Length of a 250g cylinder (D = 10 cm)?

   $L = 4m/\rho\pi D\text{^}2$

   L ≈ 0.375cm.

3. Mass of a sphere (r = 10cm)?

   m = ρ $4/3\pi r\text{^}3$

   m ≈ 35,507.41g or 35.51kg.

4. Combined data reduces standard deviation due to increased sample size.

Acknowledgements

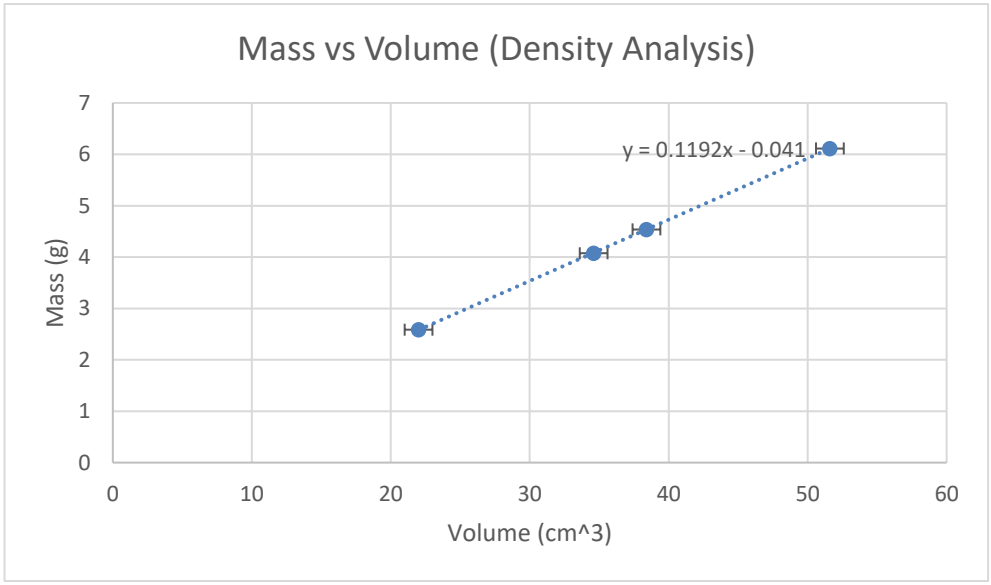
I want to thank myself for trying my best, some of the students in classroom where were helping out once they understood some of the graphing parts.

References

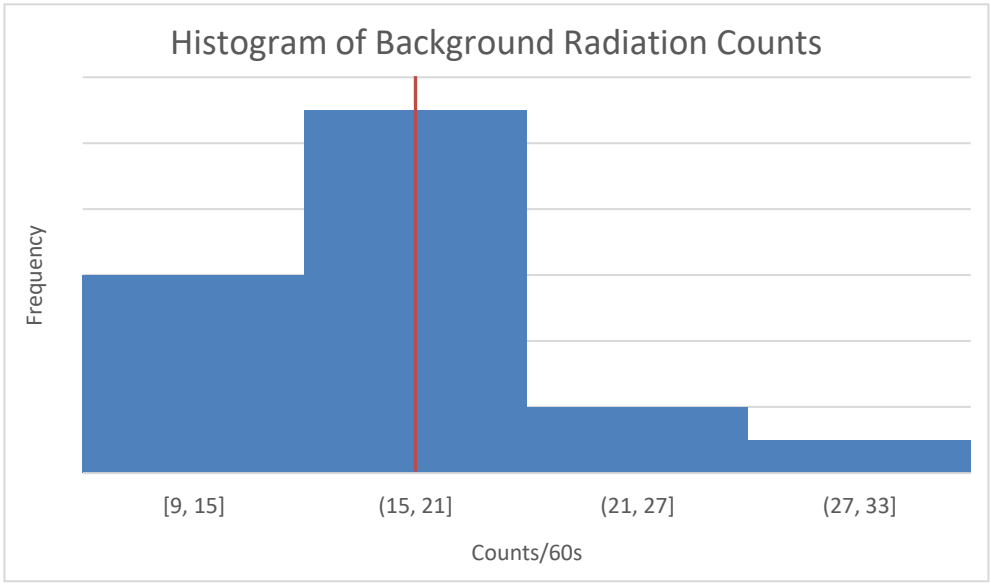
Northeastern University Physics Lab Manual (2025)

“Statistical Methods in Physics,” Journal of Measurements, 2024.

Cylinder #	#1	#2	#3	#4
Mass (m, g)	22	51.6	34.6	38.4
Uncertainty Mass (sm, g)	0.05	0.05	0.05	0.05
Relative Error (sm/m, %)	0.227273	0.096899	0.144509	0.130208
Length (L, cm)	3.65	4.85	5.7	6.35
Uncertainty Length (sL, cm)	0.05	0.05	0.05	0.05
Relative Error (sL/L, %)	1.369863	1.030928	0.877193	0.787402
Diameter (D, cm)	0.95	1.267	0.954	0.954
Uncertainty Diameter (sD, cm)	0.0005	0.0005	0.0005	0.0005
Relative Error (sD/D, %)	0.052632	0.039463	0.052411	0.052411
Volume (V, cm^3)	2.5872	6.114837	4.07438	4.539002
Uncertainty Volume (sV, cm^3)	0.035493	0.063132	0.035868	0.035898
Relative Error Volume (sV/V, %)	1.371884	1.032437	0.880319	0.790882
Density (ρ, g/cm^3)	8.503402	8.438492	8.49209	8.46001
Uncertainty Density (sρ, g/cm^3)	0.118247	0.087505	0.075758	0.067809
Relative Error Density (sρ/ρ, %)	1.390582	1.036975	0.892101	0.801529
Average Density (ρ, g/cm^3)	8.473499			
Standard Error in Density (g/cm^3)	0.08733			



Trial	Counts/60s
1	19
2	9
3	14
4	22
5	20
6	28
7	16
8	23
9	21
10	20
11	15
12	14
13	20
14	9
15	20
16	21
17	15
18	16
19	17
20	21
21	16
22	26
23	19
24	21
25	12
26	21
27	21
28	21
29	19
30	19
31	17
32	14
33	22
34	29
35	11
36	16
37	12
38	13
39	22
40	20
41	16
42	27
43	19



44	26
45	18
46	21
47	20
48	13
49	17
50	15
51	23
52	16
53	17
54	17
55	17
56	12
57	19
58	16
59	14
60	16

Mean	18.16667		
FWHM Sta	13	Max Frequency	
FWHM End	23	12	

Statistics	Value
Average Co	18.16667
Standard D	4.369256
Uncertaint	0.564068